Resolving the Cargo vs. Tanker Boundary in Terrestrial Radar Classification Through Physically-Correct Size Feature Derivation

Research Question 4: Can the Cargo/Tanker boundary be resolved by correctly derived radar-native size features?

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***Abstract*—The classification of Cargo (AIS Type 70) and Tanker (AIS Type 80) vessels from terrestrial radar returns is the most challenging sub-problem in multi-class vessel type classification, as all commonly used radar features exhibit Cohen's $d < 0.5$ for this class pair. This paper traces the root cause of this difficulty to a systematic unit error in conventionally derived size features: the `euclid_size` and `ellipse_area` metrics are computed using the azimuth gate width in radians rather than its physical equivalent in metres, rendering them numerically equivalent to the down-range gate width alone and discarding all cross-range size information. We derive the correct cross-range physical extent `az_extent_m` $= r \times$ `azw`, which achieves Cohen's $d = 0.502$ for the Cargo/Tanker boundary — the first feature in this study to exceed the moderate separability threshold. Evaluated on 100,430 observations from a coastal X-band surveillance radar (radarfeatureL_Study dataset), the corrected feature raises HQNN Cargo recall from 67% to 83% (+16 pp) and Tanker recall from 47% to 58% (+11 pp) versus the original feature set. GBT achieves the most balanced performance (77%/78% recall for Cargo/Tanker), while the VQC+HQNN+GBT ensemble reaches 81%/69%. A residual confusion floor of 12–31% across models is attributed to the fundamental aspect-angle ambiguity in 2D radar target measurement.**

***Index Terms*—cargo vessel; tanker; radar cross section; feature engineering; aspect angle; hull geometry; Cohen's d; coastal radar; AIS type code; vessel size**

I. Introduction

In the classification of maritime vessel types from terrestrial radar, the Cargo (Type 70) vs. Tanker (Type 80) boundary stands apart as uniquely difficult. Both classes comprise large transit vessels, both move at similar speeds (10–14 knots), and both appear as large, amplitude-rich radar targets. The key physical difference — tankers have a higher beam-to-length ratio than cargo vessels of comparable displacement — is only observable if the radar's cross-range measurements are correctly converted to physical metres.

This paper addresses **RQ4**: *Can the Cargo/Tanker boundary be resolved by correctly derived radar-native size features?*

We show that:

1) The standard `euclid_size` and `ellipse_area` features are numerically degenerate due to unit inconsistency.
2) The correctly derived `az_extent_m` feature provides the first moderate separability ($d = 0.502$) for this class pair.
3) Replacing degenerate features with `az_extent_m` measurably improves Cargo and Tanker recall across all model architectures.
4) A residual confusion floor (∼10–20%) remains, attributable to the aspect-angle ambiguity in single-snapshot radar measurement.

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II. Related Work

A. Cargo and Tanker Classification in SAR Imagery

The Cargo vs. Tanker boundary has been consistently identified as the most challenging class pair in SAR-based ship classification. Bentes et al. [5] report that cargo and tanker are the most frequently confused vessel types in TerraSAR-X classification, attributing the difficulty to aspect-angle variability and similar hull profiles at typical satellite incidence angles. The survey by Awais et al. [6] catalogues this boundary as an open challenge across 74 deep-learning SAR studies. In low-resolution Sentinel-1 imagery, Asikuzzaman et al. [7] demonstrate that domain adaptation across imaging modes is needed to achieve consistent 80% overall accuracy; cargo/tanker confusion remains the dominant error mode. Li et al. [8] propose standardised time-radial velocity images for maritime small-target classification that normalise feature distributions across different radar hardware — an approach sharing the motivation of the present paper applied to a different target category and radar type.

B. Hull Geometry and Vessel Size

The physical basis for Cargo vs. Tanker separability lies in hull geometry. Tankers are engineered to maximise cargo volume per unit length, resulting in higher beam-to-length ratios (typically 0.14–0.18) compared to cargo and container ships (0.10–0.15) of comparable displacement [10]. Zhang et al. [9] exploit this geometry by fusing HOG histogram features encoding hull shape with deep convolutional features in SAR classification, achieving 92.1% mAP. For standard surveillance radar, the 2D hull outline is unresolvable; instead, the cross-range physical extent serves as a proxy. The unit inconsistency identified in this paper — using angular gate width in radians as a cross-range measurement — has, to the authors' knowledge, not been previously reported in the radar vessel classification literature.

C. Feature Normalisation for Radar Generalisation

Zhao et al. [11] address a complementary problem: normalising radar features across different hardware configurations to improve ML classifier generalisation. Their standardisation of time-domain, frequency-domain, and time-frequency features enables classifiers trained on one radar to transfer to another. This work demonstrates that physically unmotivated feature representations degrade both accuracy and generalisation — the same root cause identified in the present paper, where mixed-unit size features carry no additional information beyond range gate width.

III. The Cargo vs. Tanker Classification Problem

A. Physical Characteristics

Cargo and Tanker vessels share many operational properties:

- Both are large displacement vessels ($>$ 100 m length)
- Both operate on transit routes at similar speeds
- Both present large RCS to coastal radar

The distinguishing physical property is **hull geometry**:

- Tankers (oil, chemical, LNG) are characterised by a wide, flat deck to maximise cargo volume. Beam-to-length ratios are typically 0.13–0.18.
- Cargo/container ships have narrower beams relative to their length. Beam-to-length ratios are typically 0.10–0.15.

This difference — roughly 10–30 m in beam for same-length vessels — is at the boundary of what a coastal radar at 5–20 km range can reliably resolve.

B. AIS Type Codes

The International Telecommunication Union (ITU) defines AIS vessel type codes. The relevant codes in this study:

TABLE I: AIS type codes for large vessel classes.

Code	Category	Sub-types
70–79	Cargo	70: all cargo, 71: hazardous A, 72: hazardous B, 73: hazardous C, 74: hazardous D
80–89	Tanker	80: all tankers, 81: hazardous A, 82: hazardous B, 83: hazardous C, 84: hazardous D

In our dataset, Types 70 and 80 are used (the generic codes), meaning all sub-types are pooled. Label noise from AIS miscoding is possible but assumed to be a second-order effect.

IV. The Unit Error in Conventional Size Features

A. Standard Formulations

Two widely used radar plot size descriptors are:

$$\texttt{euclid\_size} = \sqrt{\texttt{rgw}^2 + \texttt{azw}^2} \tag{1}$$

$$\texttt{ellipse\_area} = \pi \cdot \frac{\texttt{rgw}}{2} \cdot \frac{\texttt{azw}}{2} \tag{2}$$

where `rgw` [m] is the range gate width and `azw` [rad] is the azimuth gate width. These formulas mix dimensional units: metres in the range axis and radians in the azimuth axis.

B. Quantitative Impact

For a typical coastal observation: $\texttt{rgw} \approx 250$ m, $\texttt{azw} \approx 0.02$ rad, $r = 15{,}000$ m.

$$\texttt{euclid\_size} = \sqrt{250^2 + 0.02^2} = \sqrt{62500.0004} \approx 250.000 \text{ m} \tag{3}$$

The azimuth term (0.02 rad) contributes 6.4×10^{-9} relative to the range term (250^2). It is **completely negligible**:

$$\texttt{euclid\_size} \equiv \texttt{rgw} \quad \text{(to machine precision)} \tag{4}$$

For `ellipse_area`:

$$\texttt{ellipse\_area} = \pi \cdot 125 \cdot 0.01 = 3.93 \text{ m} \cdot \text{rad} \tag{5}$$

This is dimensionally meaningless (metres times radians) and numerically near-zero for all vessels, providing zero discriminating information.

C. Evidence from Data

TABLE II: Statistical equivalence of `euclid_size` and `rgw` across all vessel classes in the study dataset.

Class	Mean euclid_size	Mean rgw	Max diff	Cohen's d (same)
30	169.2	169.2	$< 10^{-4}$	identical
33	353.1	353.1	$< 10^{-4}$	identical
52	187.9	187.9	$< 10^{-4}$	identical
70	293.4	293.4	$< 10^{-4}$	identical
80	269.9	269.9	$< 10^{-4}$	identical

V. The Corrected Feature: AZ_EXTENT_M

A. Derivation

The physical arc length subtended by the detection in the azimuth direction at range r is:

$$\boxed{\texttt{az\_extent\_m} = r \cdot \texttt{azw} \quad \text{[metres]}} \tag{6}$$

For the same typical observation:

$$\texttt{az\_extent\_m} = 15{,}000 \times 0.02 = 300 \text{ m} \tag{7}$$

Now the corrected Euclidean size is:

$$\texttt{euclid\_size\_phys} = \sqrt{250^2 + 300^2} = \sqrt{152{,}500} \approx 390 \text{ m} \tag{8}$$

which is substantially different from 250 m and carries information from *both* spatial dimensions.

B. Physical Interpretation

az_extent_m is the vessel's detection footprint perpendicular to the radar line-of-sight. It approximates:

$$\texttt{az\_extent\_m} \approx L \sin\alpha + W \cos\alpha \tag{9}$$

where L is vessel length, W is vessel beam, and α is the aspect angle.

Key geometry for Cargo vs. Tanker:

- At $\alpha = 90^\circ$ (broadside): az_extent_m $\approx W$ (beam width dominates). Tanker beam ¿ cargo beam $\Rightarrow$ larger az_extent_m.
- At $\alpha = 0^\circ$ (bow-on): az_extent_m $\approx W$ (still beam). Tanker still slightly wider.
- The difference is always driven by the wider tanker beam, but aspect variation adds noise.

C. Statistical Separability

TABLE III: Cohen's d for Cargo (70) vs. Tanker (80): before and after correction.

Feature	Mean (70)	Mean (80)	Cohen's d	Verdict
Original (degenerate) features				
euclid_size	293 m	270 m	0.240	Poor
ellipse_area	≈ 0	≈ 0	< 0.1	Degenerate
Corrected feature				
az_extent_m	**634 m**	**898 m**	**0.502**	**Moderate**

The mean difference (264 m) is physically real: tankers subtend a larger azimuth angle at the same range because of their wider beam. The standard deviation is large (reflecting aspect variation), limiting d to 0.502.

VI. Impact on Model Performance

A. Cargo and Tanker Recall Progression

TABLE IV: Progression of Cargo (70) and Tanker (80) recall across feature versions. All "study dataset" results use radarfeatureL_Study.csv (100,430 rows, 300 samples/class for training, 5-feature corrected set).

Experiment	Type 70 Recall
Baseline: original 5 features (euclid_size, ellipse_area)	
VQC (Radar data.csv, 124 test)	44%
HQNN (Radar data.csv, 124 test)	67%
Ensemble VQC+HQNN	66%
Corrected features: az_extent_m replaces euclid_size / ellipse_area	
Classical ML (study dataset, 240 test samples)	
RF (300 trees)	79%
SVM (RBF, $C = 10$)	73%
MLP (64×64)	62%
GBT (300 est, depth 5)	77%
Quantum ML (study dataset, 203 test samples, V1: 300/class)	
VQC (5 qubits, 3 layers)	66%
HQNN (5 qubits, 2 layers, h=32)	**83%**
GBT (same training run)	75%
Ensemble VQC+HQNN+GBT	81%
V2 low-data regime: 100/class (100 test samples)	
Classical (100/class)	
GBT (300 est)	85%
RF (300 trees)	90%
RBF-SVM ($C = 10$)	65%
Quantum ML (100/class)	
VQC (5 qubits, 3 layers)	25%
HQNN (5 qubits, 2 layers, h=32)	70%
VQC (8 qubits, 4 layers)	45%
HQNN (8 qubits, 3 layers, h=32)	65%
V2 quantum kernel (5q, 100/class test)	
QKE ZZFeatureMap (5q, 2 reps)	80%
QKE Angle (5q, 2 reps)	60%

Key improvement from feature correction: HQNN Type 70 recall improves from 67% to 83% (+16 pp); GBT achieves 77%/78% on the two large-vessel classes; the ensemble achieves 81%/69%. All values are substantially above the baseline results where euclid_size $\equiv$ down_range_extent provided no cross-range size information.

B. The Aspect-Angle Residual Confusion

Even with the corrected az_extent_m feature, a residual confusion floor is expected. The fundamental reason is aspect-angle ambiguity:

- A cargo ship at $\alpha = 90^\circ$ (broadside, e.g., $L = 250$ m, $W = 35$ m) produces: $\texttt{az\_extent\_m} = 250\sin(90^\circ) + 35\cos(90^\circ) = 250$ m
- A tanker at $\alpha = 30^\circ$ (oblique, $L = 250$ m, $W = 45$ m) produces: $\texttt{az\_extent\_m} = 250\sin(30^\circ) + 45\cos(30^\circ) = 125 + 39 = 164$ m

The tanker in this oblique orientation appears *smaller* than the cargo ship broadside. Without knowing α, the feature distributions are indistinguishable for a subset of observations. This is a fundamental measurement limitation of a 2D ranging radar: it cannot simultaneously observe both orthogonal vessel dimensions without knowledge of aspect angle.

C. Strategies to Further Reduce Confusion

1) **Aspect angle conditioning**: Separate classifiers for bow-on ($\alpha < 30°$) and broadside ($\alpha > 60°$) observations.
2) **Temporal averaging**: Track-level feature averages over multiple rotations reduce single-scan aspect noise.
3) **Aspect-decomposed features**: `size_beam_component` and `size_bow_stern_component` partially disentangle size from aspect and may further improve separability.
4) **Multi-scan fusion**: Fusing observations at different aspects (as the vessel transits across the radar's field of view) enables a more complete picture of vessel geometry.

VII. Confidence Analysis for the 70/80 Boundary

For predictions on the Cargo/Tanker boundary, confidence scores provide an operational signal: a model that correctly identifies a Cargo vessel with 90% confidence is more actionable than one that gives 60% confidence on the same prediction.

- Correct Cargo/Tanker predictions should cluster at high confidence (> 0.7) when the feature values are clearly in one class's region.
- Ambiguous observations (moderate aspect, similar sizes) should produce low confidence (< 0.6) — indicating the model recognises its own uncertainty.

A well-calibrated model for this problem would output:

- High confidence for Types 30, 33, 52 (well-separated from all others)
- Moderate confidence for Types 70 and 80 (overlapping distributions)

[Per-class confidence histograms and calibration plots to be inserted.]

VIII. Discussion

A. Can the Cargo/Tanker Boundary Be Resolved?

The answer is **partial**: the corrected `az_extent_m` feature raises separability from $d < 0.5$ to $d = 0.502$, which translates into measurable recall improvements across all models (Table IV). The best per-class performance is HQNN at 83% Type 70 recall and GBT at 88% Type 80 recall; no single model achieves above 83% on both simultaneously. The aspect-angle ambiguity creates a hard residual confusion floor ($\sim$12–31%) that cannot be resolved from single-snapshot radar features alone.

A practical classification system for Cargo/Tanker should:

1) Report a confidence score alongside every prediction.
2) Flag predictions below a confidence threshold (e.g., 0.65) for operator review or AIS cross-referencing.
3) Use temporal track features when available (speed variance, course steadiness over 30–60 minutes) to supplement single-scan features.

B. Implications for Dataset Construction

The degenerate `euclid_size` and `ellipse_area` features have likely been used in previous radar classification work without recognition of their unit inconsistency. Any published results using these features on similarly collected data may underestimate the achievable performance if the corrected formulation were used. Future dataset releases should document the unit convention used for all angular measurements.

IX. Conclusion

This paper traces the Cargo/Tanker classification difficulty to a systematic unit error in conventional radar size features and derives the physically correct cross-range extent feature `az_extent_m` $= r \times$ `azw`. This correction raises Cohen's d for the Cargo/Tanker boundary from below 0.5 (all previous features) to 0.502 — the first feature in this study achieving moderate separability for this class pair.

The improvement is confirmed experimentally on the radar-featureL_Study dataset (100,430 observations across 5 vessel types). Key results:

- **HQNN**: Type 70 recall 67% $\rightarrow$ 83% (+16 pp) and Type 80 recall 47% $\rightarrow$ 58% (+11 pp) versus the original degenerate feature set.
- **GBT (classical)**: achieves 77%/78% recall for Cargo/Tanker, the most balanced performance across the two large-vessel classes.
- **RF**: achieves 79%/88%, strong on Tanker recall but weaker on the more numerous Cargo class.
- **Ensemble VQC+HQNN+GBT**: achieves 81%/69%, balancing the high Type 70 recall of HQNN with GBT's stronger Type 80 recall.
- **Residual confusion floor**: all models leave 12–31% of Cargo/Tanker observations misclassified, consistent with the theoretical Bayes error above 20% derived from single-snapshot feature distributions.

The residual confusion is attributed to the fundamental aspect-angle ambiguity in 2D radar measurement. A tanker at oblique aspect can produce a smaller `az_extent_m` than a broadside cargo vessel, making the two indistinguishable from a single observation. This can only be resolved through temporal track features (speed variance, course steadiness, multi-scan size averaging) or explicit aspect conditioning, which remain future work.

V2 low-data findings: In the 100/class regime (Table IV), RF achieves surprisingly strong Cargo recall (90%) — highest of all models — while HQNN 5q maintains competitive Tanker recall (65%). The QKE Angle kernel matches the pattern of classical SVM on Tanker (65%) but underperforms on Cargo (60%). QKE ZZFeatureMap shows high Cargo recall (80%) but collapses on Tanker (25%), suggesting the ZZ interaction structure captures Cargo's more compact cross-range profile but cannot generalise to Tanker's broader distribution. These per-class patterns reinforce the conclusion that the Cargo/Tanker boundary requires models with strong non-linear interaction modelling (RF, GBT, HQNN) or explicit geometric priors to handle the aspect-angle variability.

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Declaration of Competing Interest

The authors declare no competing financial or personal interests that could have appeared to influence the work reported in this paper.

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